Modeling Water Availability in Complex Terrain: A GIS-Based Water Balance and Sensitivity Analysis for Eastern Red Cedar (*Juniperus virginiana*) in Smoke Hole Gorge, West Virginia

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Abstract

This study investigates the ecohydrological conditions associated with eastern red cedar (*Juniperus virginiana*) establishment in Pendleton County, West Virginia using spatial modeling techniques. A LiDAR-derived digital elevation model (DEM) served as the foundation for calculating the Integrated Moisture Supply Index (IMSI), which integrates topographic curvature and ln-transformed flow accumulation to represent potential subsurface moisture availability. Soil Available Water Capacity (AWC) and twelve months of direct, diffuse, duration, and global solar radiation rasters were extracted for five JUVI and five OTHER reference points. A spreadsheet-based water balance model simulated monthly potential evapotranspiration (PET), actual evapotranspiration (AET), and climatic water deficit (D) under multiple AWC and IMSI scalar scenarios. Results revealed that JUVI points consistently exhibited lower IMSI values, reduced AET, and higher deficits than OTHER locations, indicating occupation of comparatively xeric microsites. Sensitivity analyses showed that changes in AWC and topographically derived IMSI multipliers significantly influenced annual AET and deficit, especially at JUVI sites. These findings align with previous studies that link cedar dominance to resource-limited environments and demonstrate the utility of GIS-based modeling approaches for evaluating forest habitat moisture gradients. The integration of topographic, edaphic, and climatic variables provides a useful framework for assessing landscape-level vegetation patterns and hydrologic response.

Introduction

Modeling environmental processes such as water balance and moisture availability is essential for understanding landscape function and supporting land management decisions. In mountainous and forested environments, climate, soil, and terrain can strongly influence local hydrology. Thus, leading to complex patterns of evapotranspiration (ET), soil moisture, and vegetation distribution. In order to quantify these interactions at landscape scales, researchers like Dyer (2019) have turned to geospatial modeling approaches that incorporate high-resolution digital elevation models (DEMs), climate data, and soil attributes.

This study builds on existing methodologies by applying a raster-based water balance model in a complex terrain setting using ArcGIS Pro. Water balance modeling was performed using monthly precipitation and potential evapotranspiration (PET) data alongside gridded estimates of available water capacity (AWC). The resulting actual evapotranspiration (AET) and moisture deficit (D) provide insight into spatial and temporal variation in water availability under different soil depth scenarios. As demonstrated by Dyer (2019), LiDAR-derived DEMs enable finer-scale modeling of hydrologic processes in topographically diverse regions, improving the accuracy of flow accumulation and related terrain derivatives.

In addition to water balance modeling, this analysis incorporates a modified version of the Integrated Moisture Supply Index (IMSI) introduced by Iverson et al. (1997). The IMSI integrates topographic curvature and upstream contributing area or flow accumulation to serve as a proxy for potential moisture convergence. This hybrid terrain-based index has shown promise in predicting forest composition and productivity by capturing long-term spatial patterns of water availability, especially where soil and climate data are sparse (Iverson et al., 1997).

To support solar energy estimation, the study utilizes monthly solar radiation rasters generated from the upward-looking hemispherical viewshed and solar azimuth model developed by Fu, P., & Rich, P. M. (2002). This physically based model calculates solar radiation in watt-hours per square meter ($W \cdot h \cdot m^{-2}$) and accounts for terrain-induced shading, slope, aspect, elevation, atmospheric transmittance, and latitude-specific variations in solar angle and azimuth (Fu, P., & Rich, P. M., 2002). One of its key strengths is its ability to simulate global, direct, diffuse, and duration-based solar radiation components, providing a picture of how solar energy interacts with complex topography, an essential factor when estimating PET and assessing vegetation microclimates (Fu, P., & Rich, P. M., 2002). Although this method is computationally intensive, the outputs were pre-generated for this study using ArcGIS's default settings to save time.

Finally, the project evaluates these terrain and climate-driven outputs across multiple modeling scenarios. These include varying soil depths and IMSI Scalars to simulate sensitivity under different climate and soil conditions. The ultimate objective is to assess how spatial variability in hydrologic inputs and terrain characteristics affects water availability, and to demonstrate how GIS-based tools can enhance environmental decision-making in mountainous regions of West Virginia.

Materials and Methods

30-Meter USGS NED DEM

The 30-meter National Elevation Dataset (NED) digital elevation model was acquired from the United States Geological Survey. This raster was originally compiled from high-resolution elevation sources and standardized to a consistent 30-meter resolution (U.S.

Geological Survey, 2010). It provides continuous elevation coverage across Pendleton County, WV, and served as the primary topographic surface for all terrain-based modeling within the study. The DEM was used to generate derivatives such as slope, curvature, flow direction, and flow accumulation inputs. These derivatives were utilized to calculate the modified IMSI and served as a baseline for simulating solar radiation patterns across the landscape.

Soil Available Water Capacity (AWC) Rasters

Four (4) AWC rasters, awc_025wta, awc_050wta, awc_0100wta, and awc_0150wta, were utilized to represent the spatial distribution of plant-available soil moisture across the landscape. These rasters were derived from NRCS SSURGO soil survey data for Pendleton County, WV, summarized in centimeters and converted to raster format using ArcGIS by West Virginia University (U.S. Department of Agriculture, Natural Resources Conservation Service, 2010a, 2010b, 2010c, 2010d). Each raster represents a different soil depth (25 cm, 50 cm, 100 cm, and 150 cm, respectively), offering multiple perspectives on rooting zone water availability. AWC values were used as an input to the water balance equation to estimate AET, allowing the model to account for spatial variability in soil moisture storage. Since soil data were used in raster form, values were directly comparable with elevation and climate rasters during raster calculator operations.

Monthly Diffuse, Direct, Duration, & Global Solar Radiation Rasters

A total of forty-eight (48) monthly solar radiation rasters were pre-generated using the Area Solar Radiation tool in ArcGIS Pro, based on the model developed by Fu, P., & Rich, P. M. (2002). These rasters represent four (4) solar radiation components, direct, diffuse, duration, and

global solar radiation across each month of the year. Datasets were generated using a LiDAR-based DEM for the Smoke Hole, WV area and provided by West Virginia University (Dyer, 2019a, 2019b, 2019c, 2019d). Inputs were processed using default parameters and account for slope, aspect, elevation, and atmospheric transmittance, consistent with the methodology outlined in Esri's documentation (Esri, 2024). Together, these data enabled estimation of spatially distributed PET based on incoming solar energy. Forming a critical input to the water balance model, which was implemented using a spreadsheet developed by McNeil, B. E. (2019).

Tree Type Points

Two (2) point datasets were used to compare vegetation moisture conditions between tree types. The first being five (5) "JUVI" points representing old-growth eastern red cedar (Juniperus virginiana), and secondly five (5) "OTHER" points used for comparison against the old-growth eastern red cedar points. The JUVI points were collected by Dr. Amy Hessl and her students during fieldwork in Pendleton County, WV (West Virginia University, 2024a; West Virginia University, 2024b). These high-confidence locations reflect distinct forest microhabitats and were used to extract modeled PET, AET, and deficit values for comparative analysis.

PRISM 30-Year Climate Normals

The PRISM 30-Year Climate Normals were used to provide monthly precipitation and temperature estimates for the Smoke Hole Gorge study area. These data were acquired from the PRISM Climate Group and reflect long-term averages based on a regression-based interpolation that integrates elevation, station data, and expert knowledge to account for orographic effects (PRISM Climate Group, 2022). PRISM-derived values were treated as spatially uniform across

the study extent and inserted into the Excel-based water balance model to represent atmospheric demand and supply conditions. These inputs enabled calculation of annual AET and water deficit (D) under assumed average climate conditions.

Water Balance Calculation Spreadsheet

A spreadsheet-based water balance model developed by Dr. Brenden McNeil at West Virginia University was used to simulate monthly PET, AET, and deficit values under different weighting scenarios for the temperature and solar radiation multipliers (McNeil, 2019). Graphs generated from these outputs, included in Appendix A, depict how water balance dynamics vary seasonally under different assumptions of solar and temperature influence. These simulations provided the basis for assessing sensitivity to PET input weights and evaluating hydrological differences between JUVI and OTHER points.

Spatial Analyses

All spatial analyses for this study were conducted using ArcGIS Pro 3.4.2 (ArcGIS, 2024). The primary objective of this study was to examine methods for assessing spatial variability in water availability and evaluate how water availability can structure plant communities within the Smoke Hole region of Pendleton County, WV. A combination of topographic, soil, climate, and solar radiation variables were used to calculate key indices and water balance metrics that influence forest moisture regimes and vegetation patterns.

The first set of analyses involved preprocessing a 30-meter resolution NED DEM to derive terrain-based inputs for the IMSI, an index adapted from the Integrated Moisture Index (Iverson et al., 1997). Terrain derivatives including flow direction, flow accumulation, and

curvature were generated using the Spatial Analyst toolbox. Flow accumulation was log-transformed and both it and curvature were normalized to a common scale. A weighted average was then computed using the Raster Calculator to produce a final IMSI raster. This approach follows Iverson et al. (1997), who demonstrated that integrating terrain position, flow routing, and convexity yields a strong predictor of relative moisture availability in forested landscapes.

Soil moisture availability was represented through a series of four (4) AWC rasters derived from SSURGO soil data at rooting depths of 25, 50, 100, and 150 cm. These rasters were processed using default ArcGIS settings and served as proxies for plant-available water storage across the study area (U.S. Department of Agriculture, Natural Resources Conservation Service, 2010a, 2010b, 2010c, 2010d). Values were incorporated into water balance calculations to spatially simulate the impact of rooting depth on AET.

Forty-eight (48) solar radiation rasters were also included, consisting of monthly diffuse, direct, duration, and global radiation outputs. These were precomputed using the Area Solar Radiation tool in ArcGIS, applying the geometric solar model of Fu and Rich (2002) to a LiDAR-based DEM for the Smoke Hole region (Dyer, 2019a, 2019b, 2019c, 2019d). The rasters accounted for topographic shading and elevation to simulate incoming solar energy on a monthly basis. The inclusion of these datasets was essential for estimating PET, which is highly sensitive to solar input (Esri, 2024).

Point-based analyses were conducted at 10 total field locations, split evenly between eastern red cedar sites and comparison forest sites (West Virginia University, 2024a, 2024b). Raster values of the twelve (12) global radiation rasters, IMSI, and the four (4) soil AWC rasters were extracted for each point using the “Sample” tool. These values were used in a spreadsheet-based water balance model developed by Dr. Brenden McNeil (West Virginia University, 2024c),

which simulated monthly hydrologic dynamics under different IMSI Scalar scenarios. Each scenario adjusted the influence of solar radiation and temperature on PET estimation, allowing for sensitivity analysis and inter-site comparisons.

By combining terrain, soil, radiation, and point-based sampling data, the spatial analyses enabled a robust assessment of landscape moisture regimes and how forest type influences and responds to hydrologic variability. This integrative, GIS-driven approach builds on previous work in forest moisture modeling (Iverson et al., 1997) and highlights the utility of spatial modeling tools for ecosystem-level analyses.

Results

Spatially explicit input data, including twelve (12) monthly global radiation rasters, IMSI, and four (4) soil AWC rasters were extracted for five (5) eastern red cedar (JUVI) points and five (5) comparison points (OTHER). Table 1 summarizes differences in these key spatial inputs. On average, JUVI points received 111,000.23 Wh/m² of global solar radiation annually, while OTHER points received 104,239.97 Wh/m². Mean IMSI values were notably lower at JUVI locations (21) than at OTHER locations (40), suggesting lower surface and subsurface moisture convergence. AWC values were also lower at JUVI sites, averaging 5.39 mm compared to 6.46 mm at OTHER points, indicating more limited rooting zone moisture capacity for eastern red cedar.

Water balance modeling results revealed differences in annual AET and deficit (D) across tree types and IMSI scalar values. As shown in Table 2, JUVI sites had higher AET than OTHER sites under both IMSI scalar conditions. For an IMSI scalar of 0.0020, JUVI AET averaged 643 mm, with a deficit of 71 mm. OTHER sites, in contrast, had AET of 626 mm and a deficit of 59

mm. At the reduced IMSI scalar of 0.0015, JUVI sites still had slightly higher AET (648 mm) and lower deficit (66 mm) compared to OTHER sites (629 mm AET, 56 mm deficit). These modest but consistent differences are summarized in Table 3. Which shows ΔAET values of +17 to +19 mm and ΔD values of +10 to +12 mm for JUVI compared to OTHER points. This equates to roughly 2.0–2.3% of annual precipitation for AET and 1.2–1.4% for D.

Sensitivity analysis across AWC values (25–150 cm) revealed nonlinear increases in AET and corresponding decreases in deficit (D), calculated as the difference between PET and AET. In Figure A1, JUVI sites under the 25 mm AWC (0.0020 IMSI Scalar) scenario showed a total AET of approximately 100 mm, while Figure A7, 150 mm AWC (0.0020 IMSI Scalar) scenario shows AET near 110 mm, representing a 10 mm increase. PET remained around 90 mm in both cases, meaning deficit decreased from roughly 27 mm to 17 mm, a 10 mm reduction.

For OTHER sites, Figure A2 25 mm AWC (0.0020 IMSI Scalar) indicates an AET of about 98 mm, increasing to nearly 114 mm in Figure A8, 150 mm AWC (0.0020 IMSI Scalar), a 16 mm gain. Deficit fell from approximately 28 mm to 13 mm, based on a constant PET around 127 mm. These proportional changes reflect the influence of soil water holding capacity on vegetation water availability.

Modeling with the new 0.0015 IMSI Scalar inputs further increased AET and reduced deficit (D). In Figure A15 AET reached approximately 112 mm, while PET remained close to 127 mm, reducing the deficit to about 15 mm. For OTHER sites, Figure A16 shows AET peaking around 115 mm, with PET remaining close to 127 mm, resulting in an estimated deficit close to 12 mm. These outcomes suggest that drier, cedar-dominated microsites (JUVI) respond more strongly to increases in simulated subsurface moisture availability.

Across all models and weighting assumptions, eastern red cedar (JUVI) points consistently maintained lower IMSI and AWC values, and experienced relatively greater changes in deficit when moisture supply increased. Collectively, these findings support the hypothesis that JUVI occupies drier microsites but responds strongly to additional moisture inputs, a conclusion clearly illustrated in the full set of seasonal water balance curves in Appendix A.

Discussion

This study highlights the value of integrating topographic, climatic, and soil variables in spatially explicit models to better understand fine-scale moisture and energy dynamics across forested landscapes. As shown in Figures A1–A16 and Table 1, soil water capacity and solar radiation have measurable impacts on ET and moisture deficit (D). Though their effects vary by location and vegetation type. At JUVI sites, for example, increasing AWC consistently raised AET and lowered deficit (D). Especially under low IMSI scalar conditions, indicating that shallow-rooted *Juniperus virginiana* may benefit from improved water storage under drier microsites. This aligns with Iverson et al. (1997), who found that soil curvature and flow accumulation metrics can be strong predictors of drought-tolerant species distribution, particularly when integrated into moisture indices like the IMSI.

Solar radiation effects were more variable, but showed expected seasonal trends. As Fu and Rich (2002) demonstrated, terrain derived variation in incoming radiation strongly influences localized microclimates and soil temperatures. In this study, higher direct radiation in summer months likely elevated PET and enhanced deficits in exposed areas, especially at OTHER sites with higher insolation exposure. These patterns highlights Dyer's (2019)

conclusion that topographically-driven radiation and water balance modeling can successfully predict vegetation-environment relationships even at fine scales.

Finally, differences in the magnitude and responsiveness of deficit between JUVI and OTHER locations suggest distinct ecological strategies. Deep-rooted species at OTHER sites maintained higher AET across scenarios, while JUVI points were more sensitive to small changes in soil or energy supply. Such spatial variability reinforces the need to pair species-specific traits with high-resolution environmental data when modeling forest response under current and future climate regimes (Iverson et al., 1999).

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Table 1. Mean values of spatial model inputs for JUVI and OTHER tree types based on GIS-derived variables. Values reflect mean solar radiation, Integrated Moisture Supply Index (IMSI), and available water capacity (AWC) across five representative points for each tree type.

Mean Model Input Values for JUVI and OTHER Tree Types

Tree Type	Mean Solar Radiation (Wh/m^2)	Mean IMSI	Mean AWC (mm)
JUVI	111,000.23	21	5.39
OTHERS	104,239.97	40	6.46

Values represent the mean of five locations per tree type. Solar radiation is reported as total annual Wh/m^2, while IMSI and AWC are unitless and mm, respectively. These averages reflect the spatial variation in water and energy availability across the landscape for eastern red cedar (JUVI) and other vegetation types.

Table 2. Mean model outputs of actual evapotranspiration (AET) and moisture deficit (D) for JUVI and OTHER tree types across the two IMSI scalar values.

Mean Annual AET and Deficit (D) for JUVI and OTHER Tree Types			
IMSI Scalar	**Tree Type**	**AET (mm)**	**D (mm)**
0.0020	JUVI	643	71
0.0020	OTHERS	626	59
0.0015	JUVI	648	66
0.0015	OTHERS	629	56

Annual AET and D values were computed as the mean of monthly outputs for each group.

Table 3. Differences in mean annual actual evapotranspiration (AET) and moisture deficit (D) between JUVI and OTHER tree types across two IMSI scalar values.

Mean Annual Differences in AET and Deficit (D) Between JUVI and OTHER Tree Types				
IMSI Scalar	**ΔAET (mm)**	**ΔAET as % of P**	**ΔD (mm)**	**ΔD as % of P**
0.0020	17	2.0%	12	1.4%
0.0015	19	2.3%	10	1.2%

Annual AET and D values were computed as the average of monthly outputs. Δ values represent JUVI minus OTHER tree type values at each IMSI scalar level. Percentages are relative to the total annual precipitation (P = 835 mm).

Appendix A.

Sensitivity Analysis - Water Balance Graphs

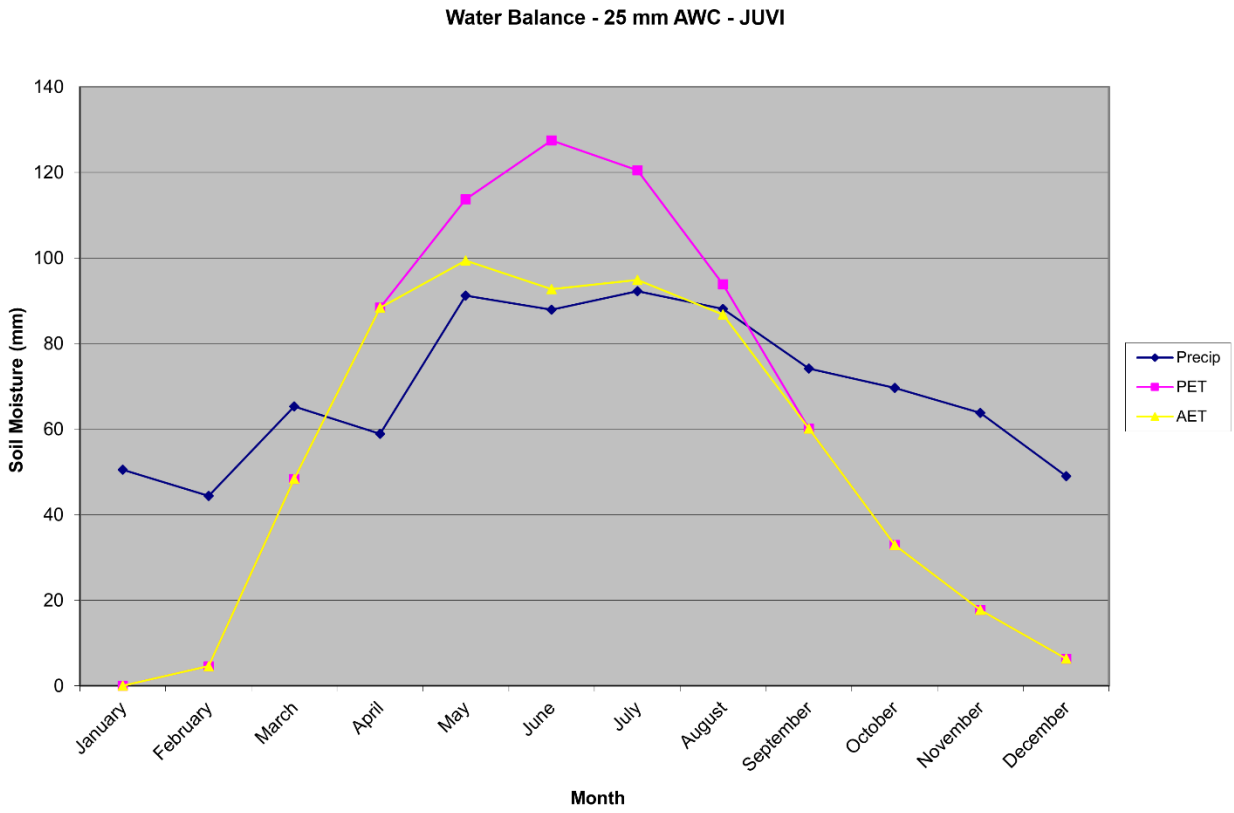

Figure A1. Graph showing monthly actual evapotranspiration (AET), potential evapotranspiration (PET), and precipitation (Precip) for JUVI under the Original model using the 25 mm soil available water capacity (AWC).

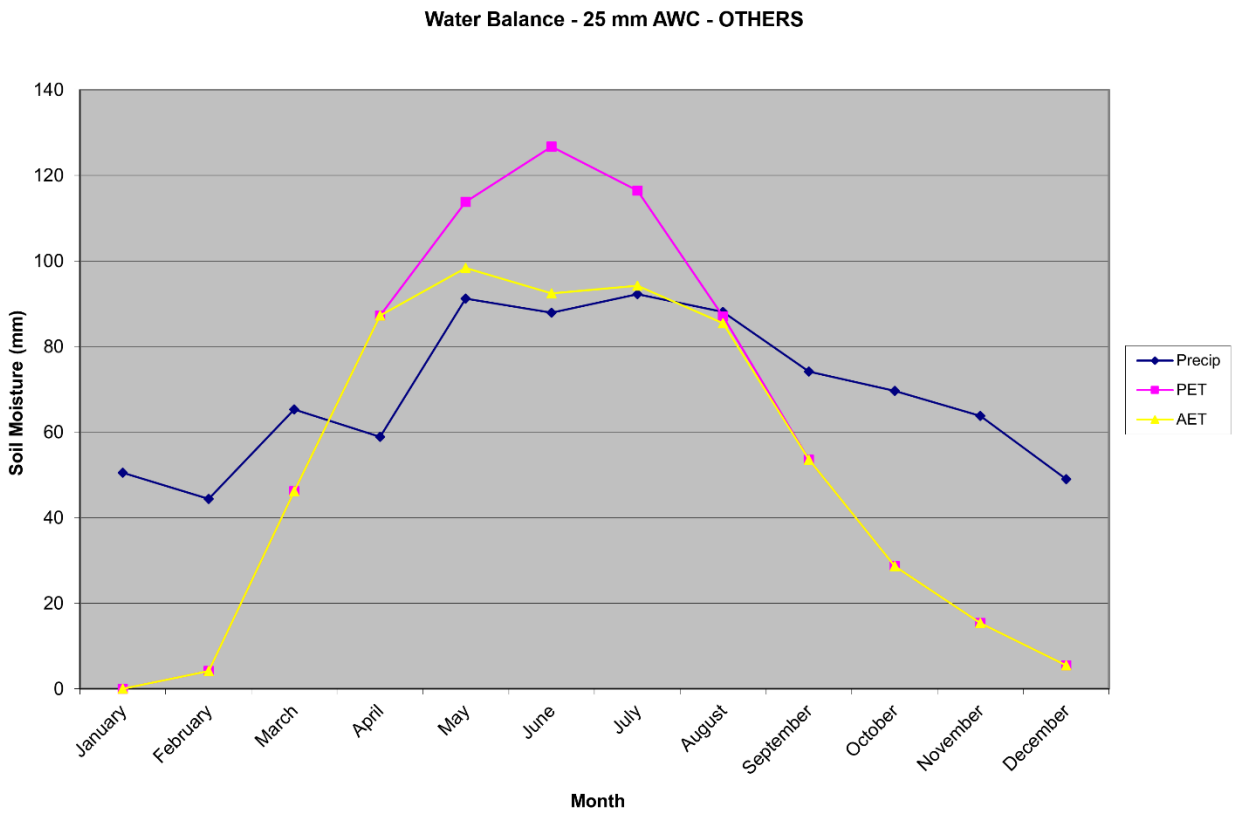

Figure A2. Graph showing monthly actual evapotranspiration (AET), potential evapotranspiration (PET), and precipitation (Precip) for OTHERS under the Original model using the 25 mm soil available water capacity (AWC).

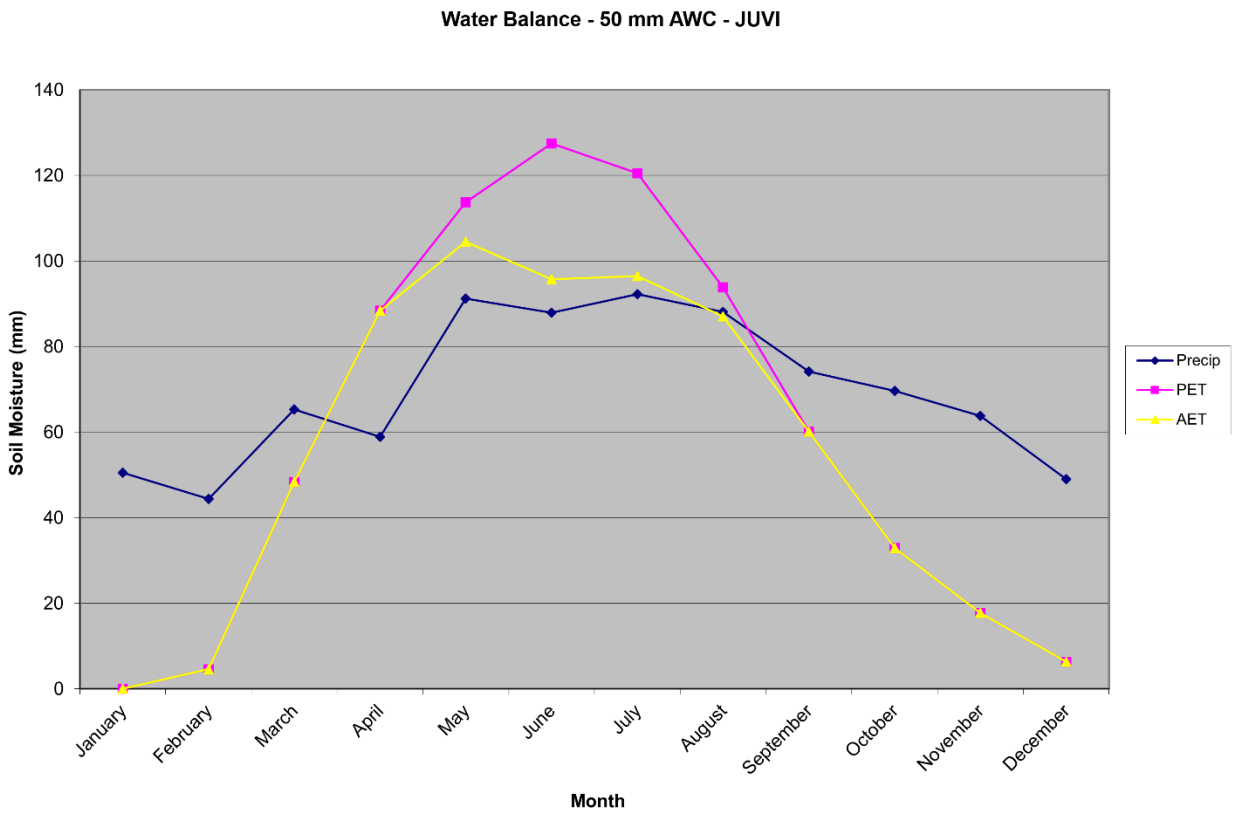

Figure A3. Graph showing monthly actual evapotranspiration (AET), potential evapotranspiration (PET), and precipitation (Precip) for JUVI under the Original model using the 50 mm soil available water capacity (AWC).

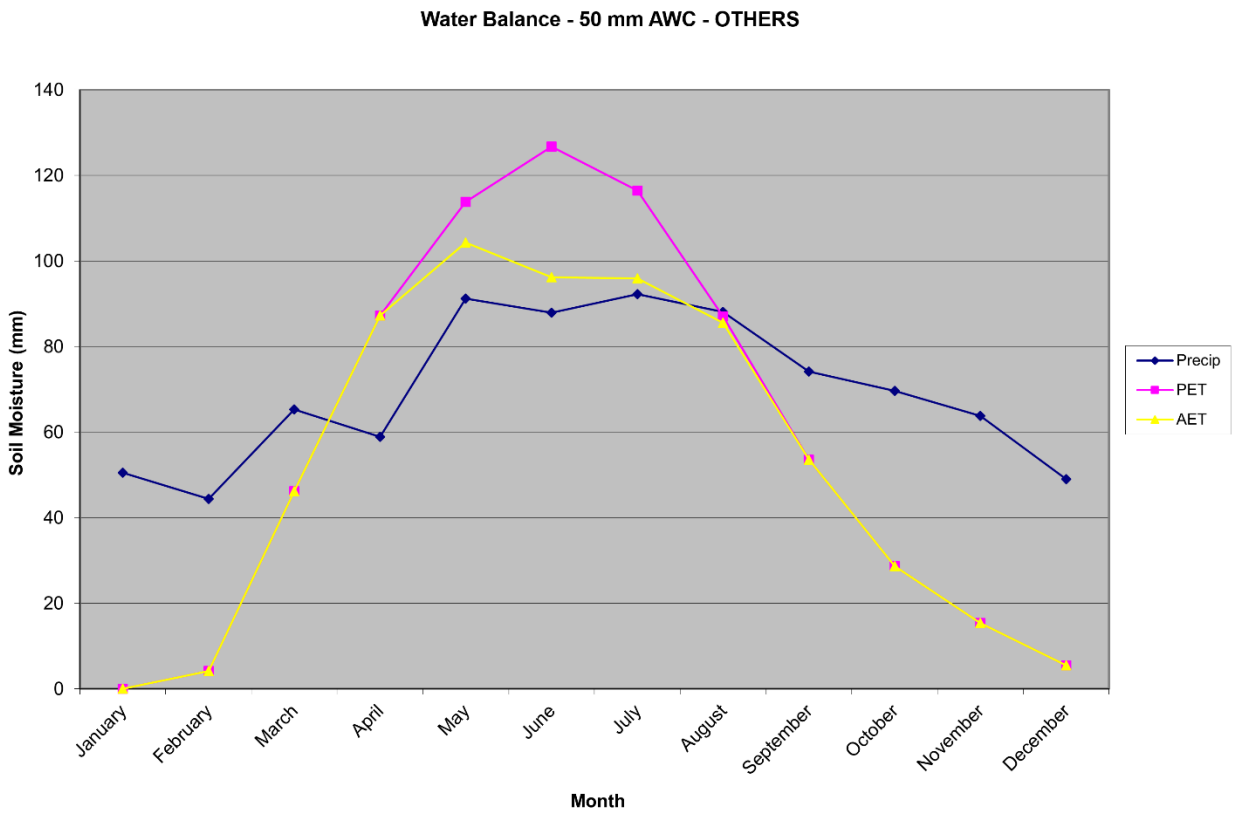

Figure A4. Graph showing monthly actual evapotranspiration (AET), potential evapotranspiration (PET), and precipitation (Precip) for OTHERS under the Original model using the 50 mm soil available water capacity (AWC).

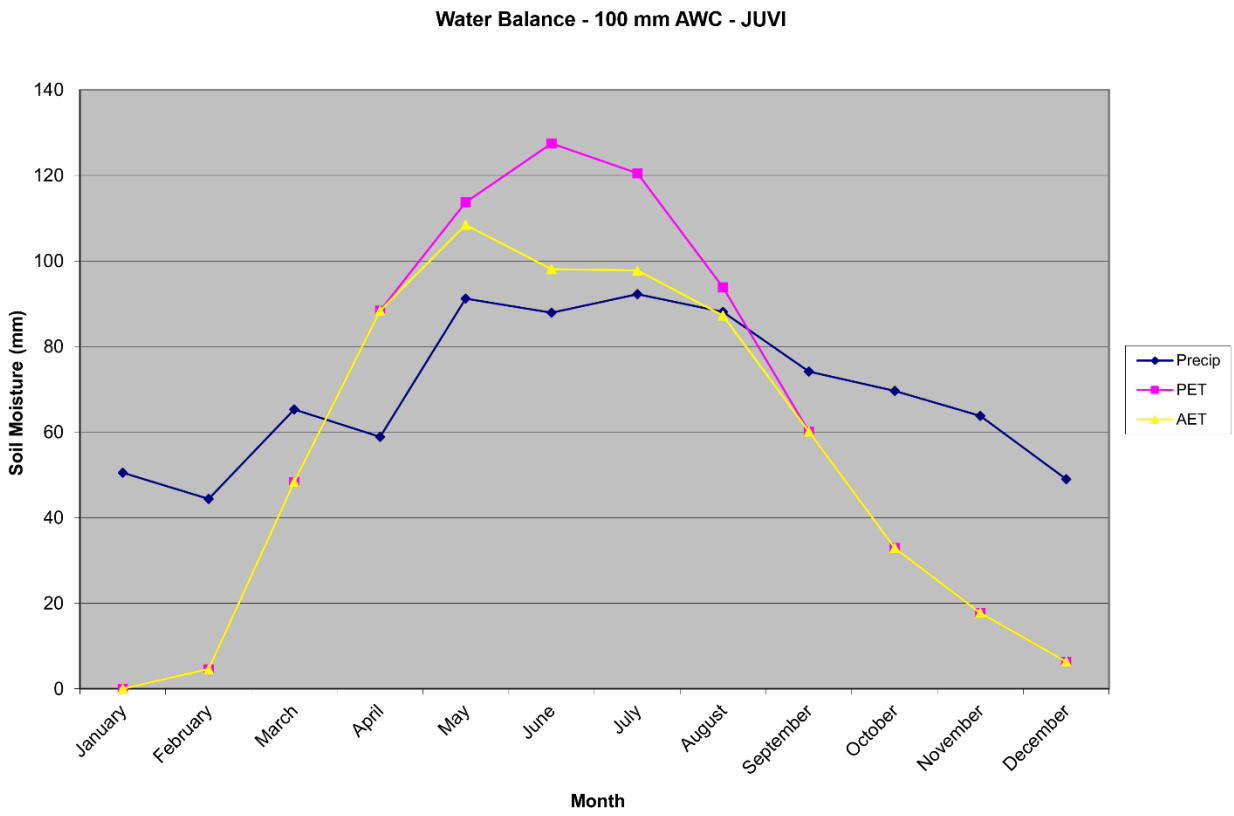

Figure A5. Graph showing monthly actual evapotranspiration (AET), potential evapotranspiration (PET), and precipitation (Precip) for JUVI under the Original model using the 100 mm soil available water capacity (AWC).

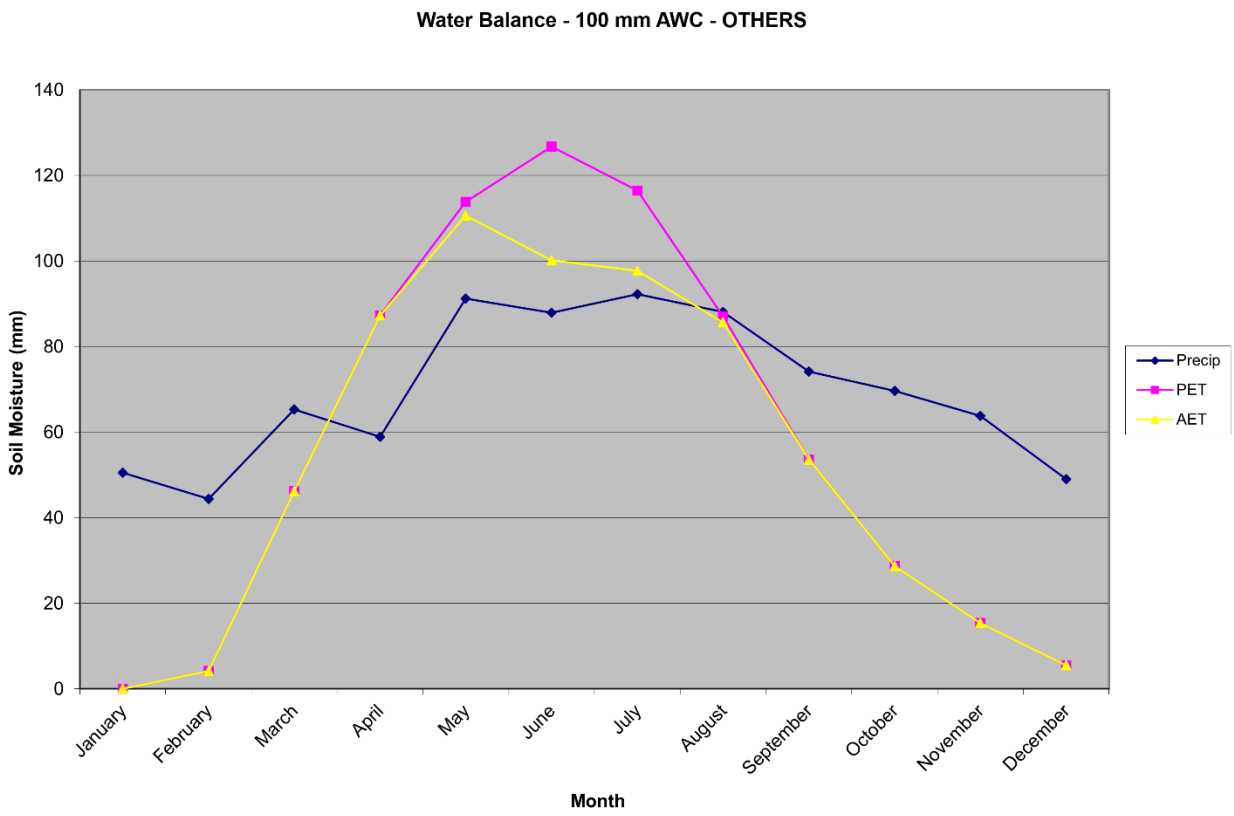

Figure A6. Graph showing monthly actual evapotranspiration (AET), potential evapotranspiration (PET), and precipitation (Precip) for OTHERS under the Original model using the 100 mm soil available water capacity (AWC).

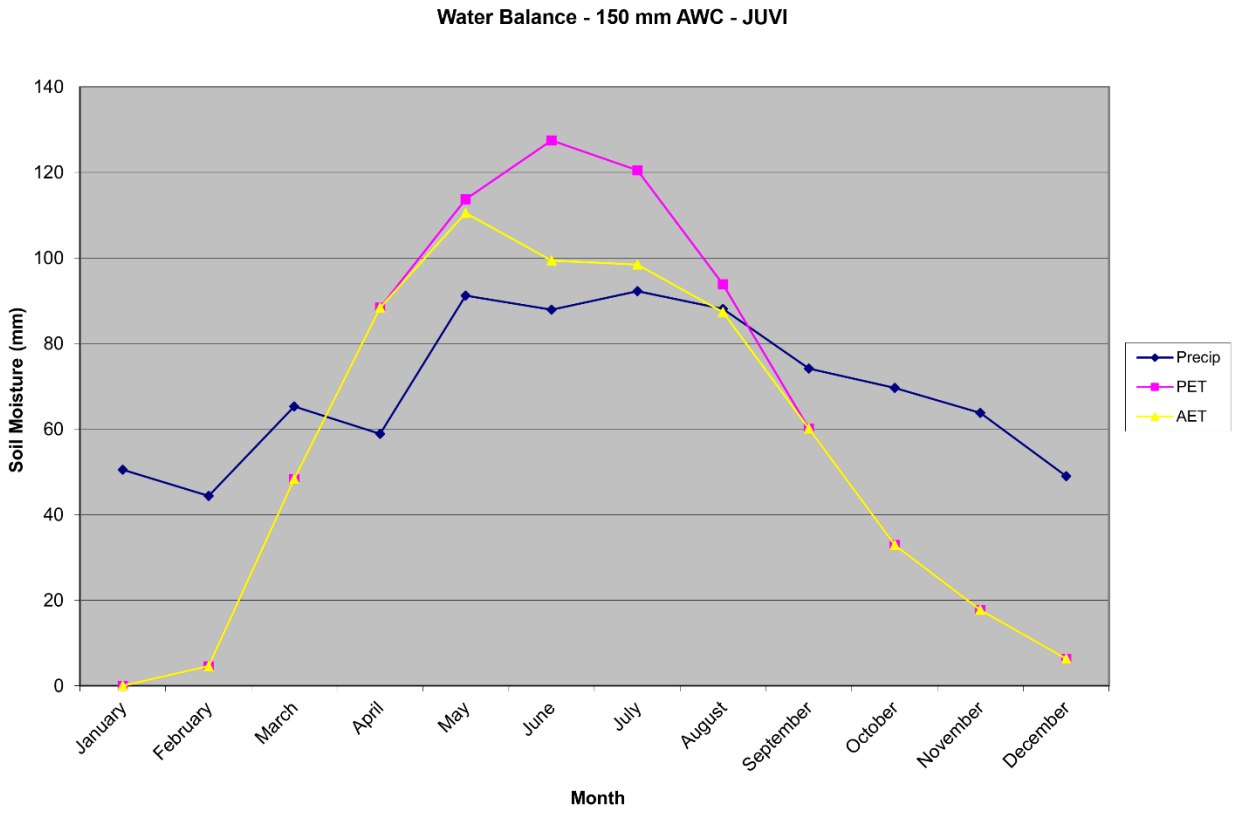

Figure A7. Graph showing monthly actual evapotranspiration (AET), potential evapotranspiration (PET), and precipitation (Precip) for JUVI under the Original model using the 150 mm soil available water capacity (AWC).

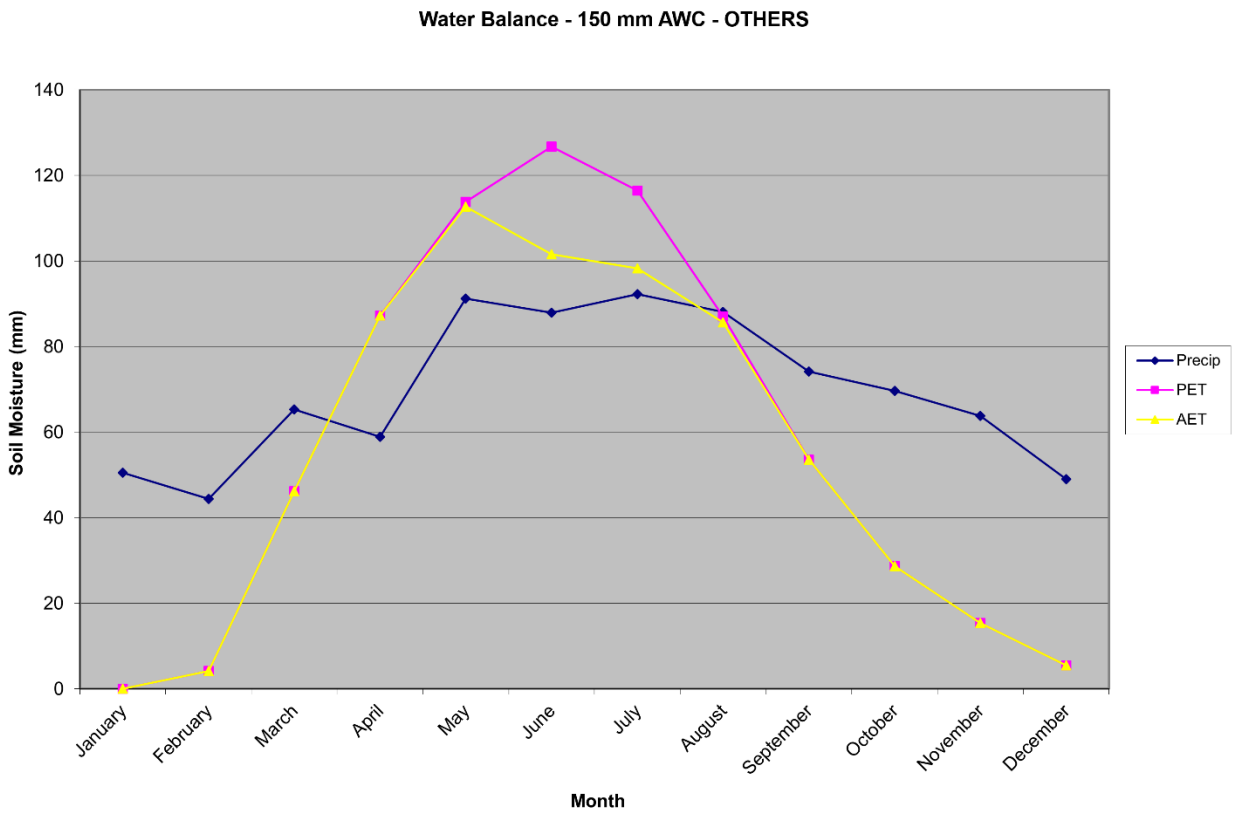

Figure A8. Graph showing monthly actual evapotranspiration (AET), potential evapotranspiration (PET), and precipitation (Precip) for OTHERS under the Original model using the 150 mm soil available water capacity (AWC).

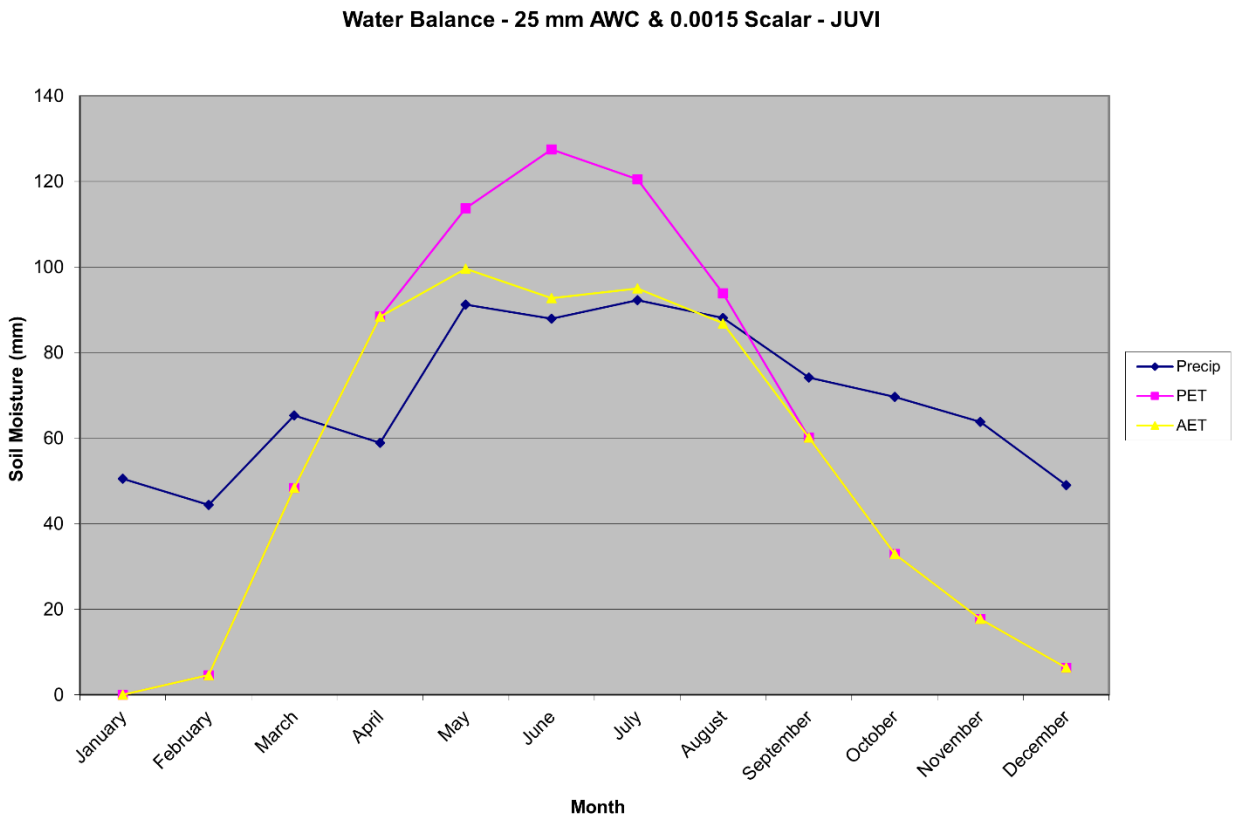

Figure A9. Graph showing monthly actual evapotranspiration (AET), potential evapotranspiration (PET), and precipitation (Precip) for JUVI under the New multiplier scenario using an IMSI Scalar of 0.0015 in lieu of 0.0020 with the 25 mm soil available water capacity (AWC).

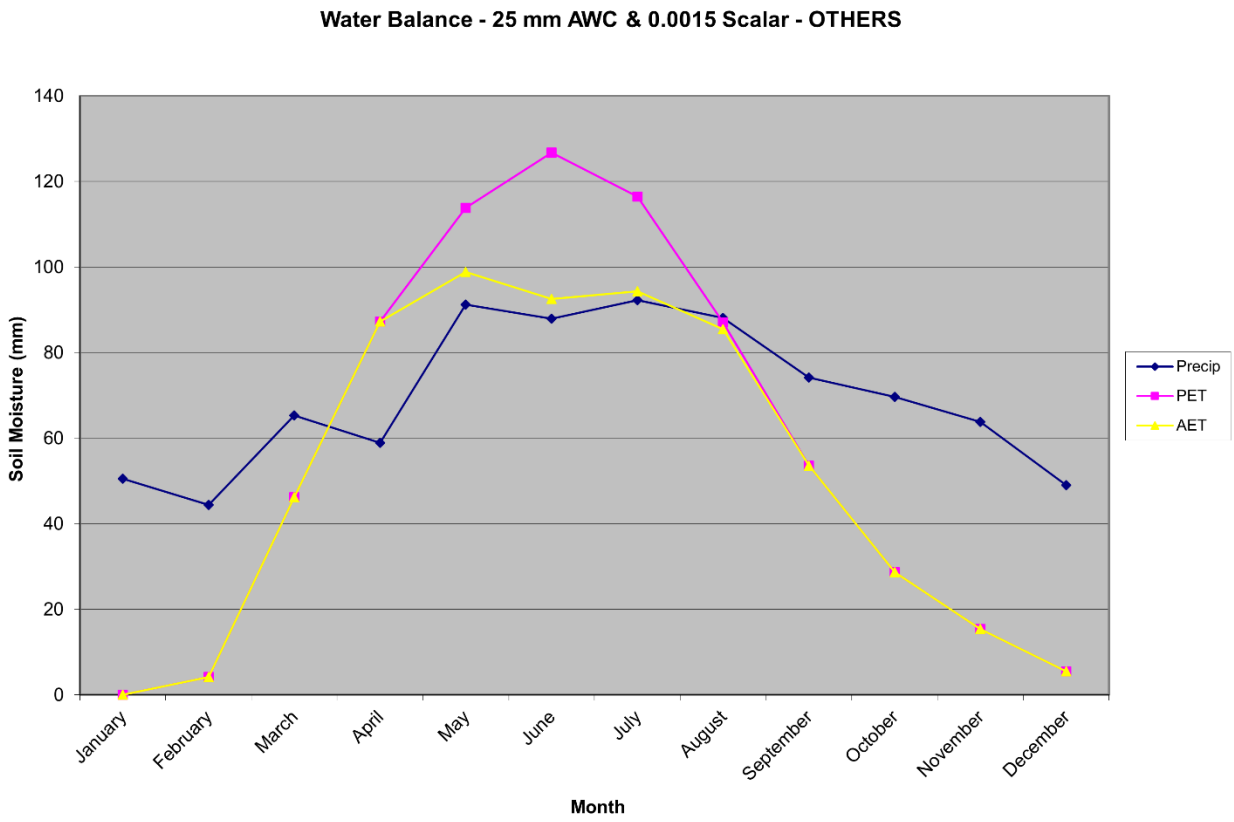

Figure A10. Graph showing monthly actual evapotranspiration (AET), potential evapotranspiration (PET), and precipitation (Precip) for OTHERS under the New multiplier scenario using an IMSI Scalar of 0.0015 in lieu of 0.0020 with the 25 mm soil available water capacity (AWC).

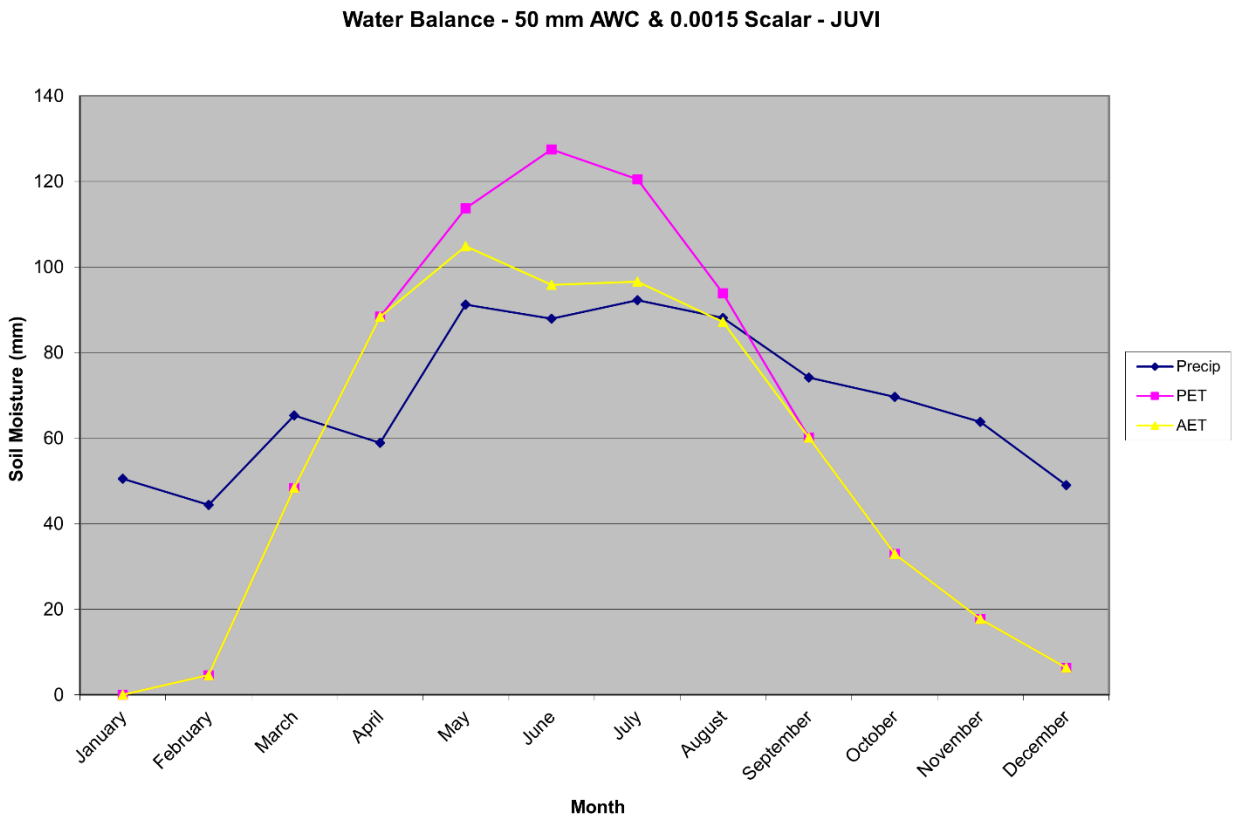

Figure A11. Graph showing monthly actual evapotranspiration (AET), potential evapotranspiration (PET), and precipitation (Precip) for JUVI under the New multiplier scenario using an IMSI Scalar of 0.0015 in lieu of 0.0020 with the 50 mm soil available water capacity (AWC).

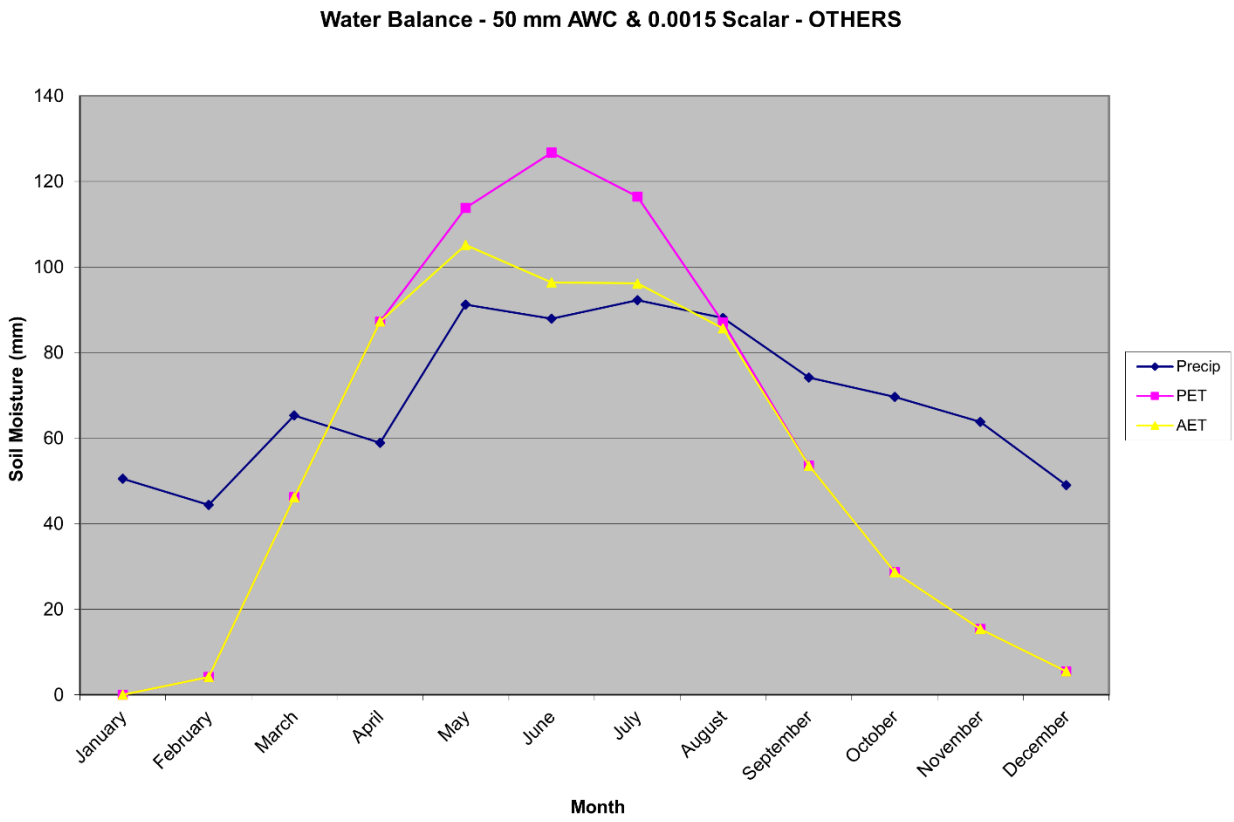

Figure A12. Graph showing monthly actual evapotranspiration (AET), potential evapotranspiration (PET), and precipitation (Precip) for OTHERS under the New multiplier scenario using an IMSI Scalar of 0.0015 in lieu of 0.0020 with the 50 mm soil available water capacity (AWC).

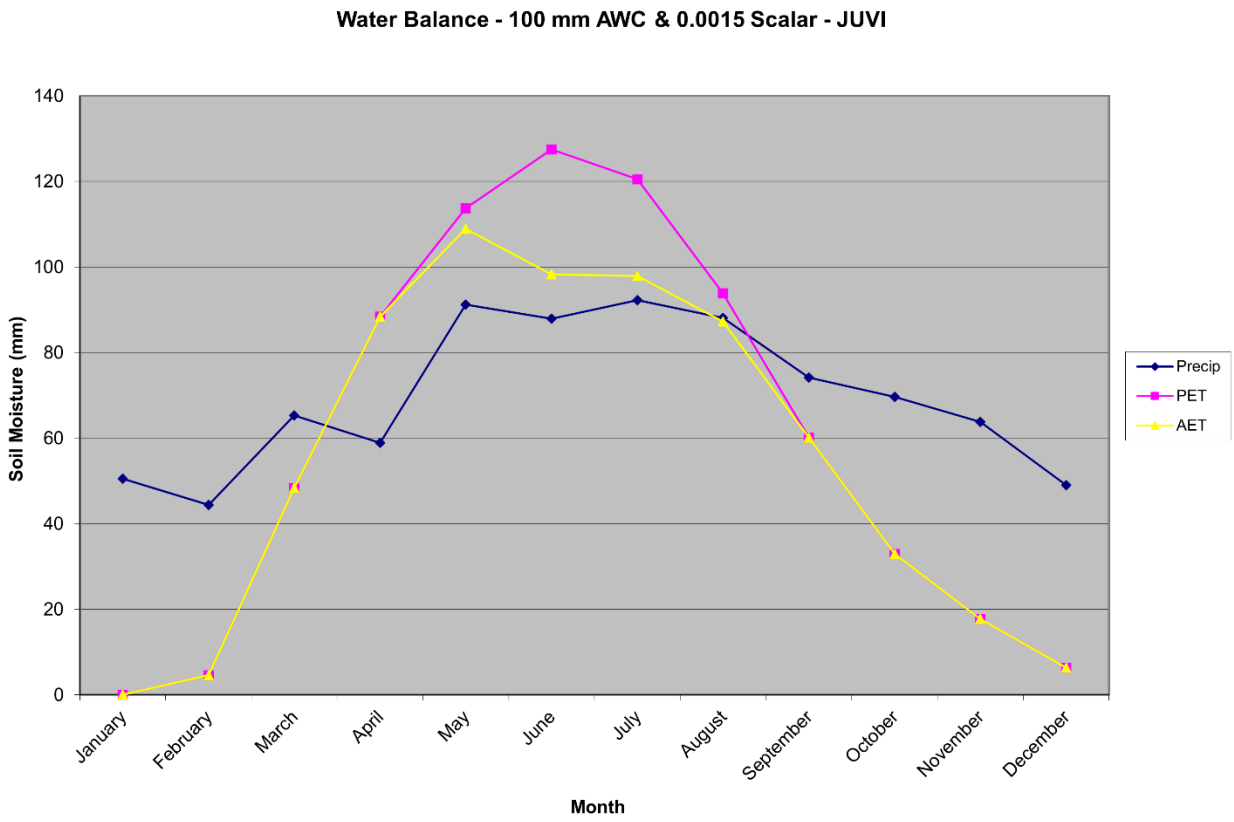

Figure A13. Graph showing monthly actual evapotranspiration (AET), potential evapotranspiration (PET), and precipitation (Precip) for JUVI under the New multiplier scenario using an IMSI Scalar of 0.0015 in lieu of 0.0020 with the 100 mm soil available water capacity (AWC).

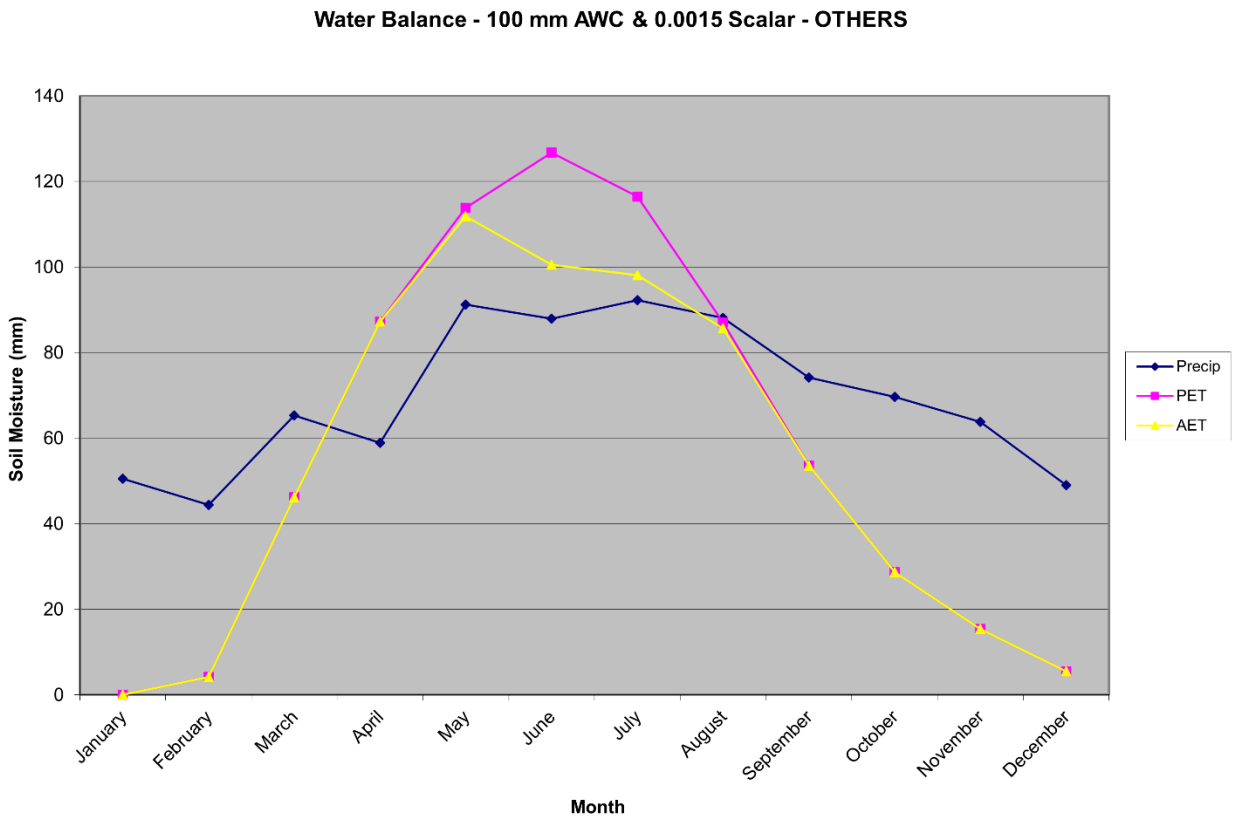

Figure A14. Graph showing monthly actual evapotranspiration (AET), potential evapotranspiration (PET), and precipitation (Precip) for OTHERS under the New multiplier scenario using an IMSI Scalar of 0.0015 in lieu of 0.0020 with the 100 mm soil available water capacity (AWC).

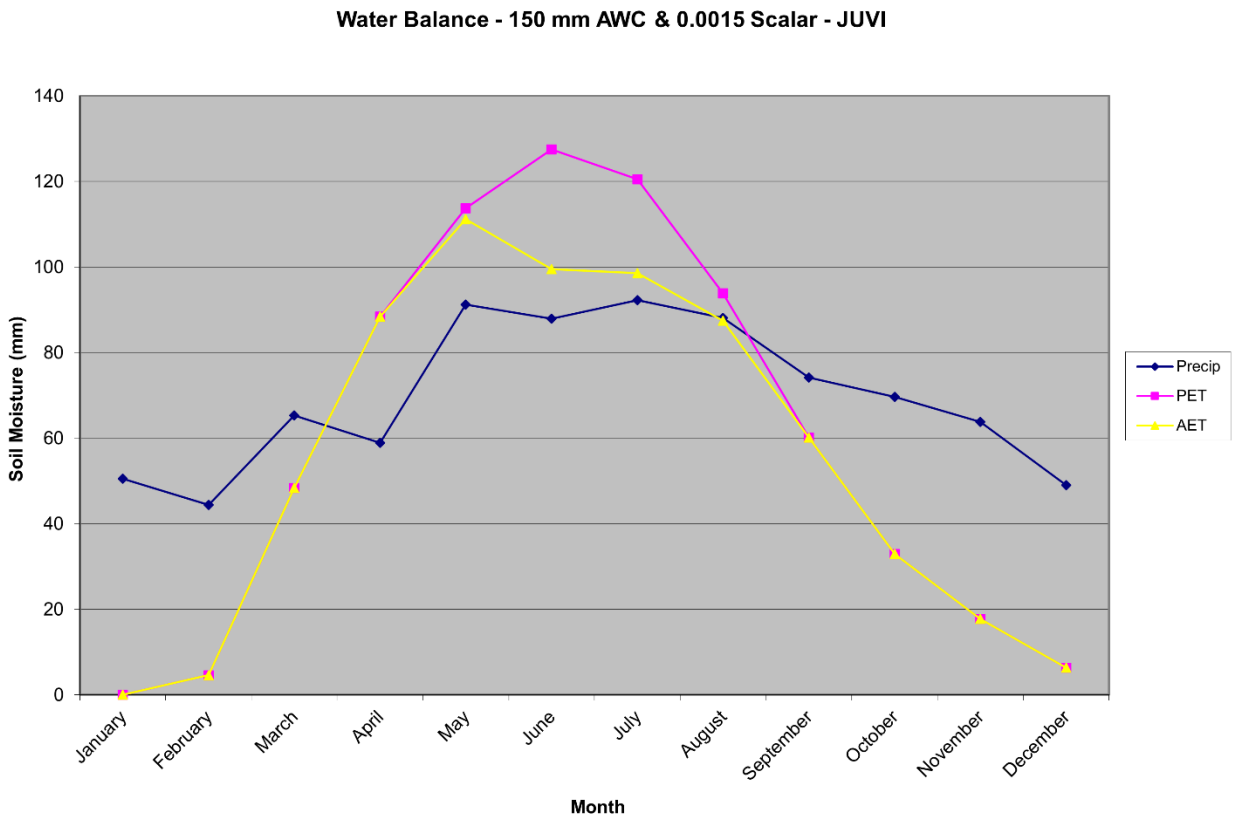

Figure A15. Graph showing monthly actual evapotranspiration (AET), potential evapotranspiration (PET), and precipitation (Precip) for JUVI under the New multiplier scenario using an IMSI Scalar of 0.0015 in lieu of 0.0020 with the 150 mm soil available water capacity (AWC).

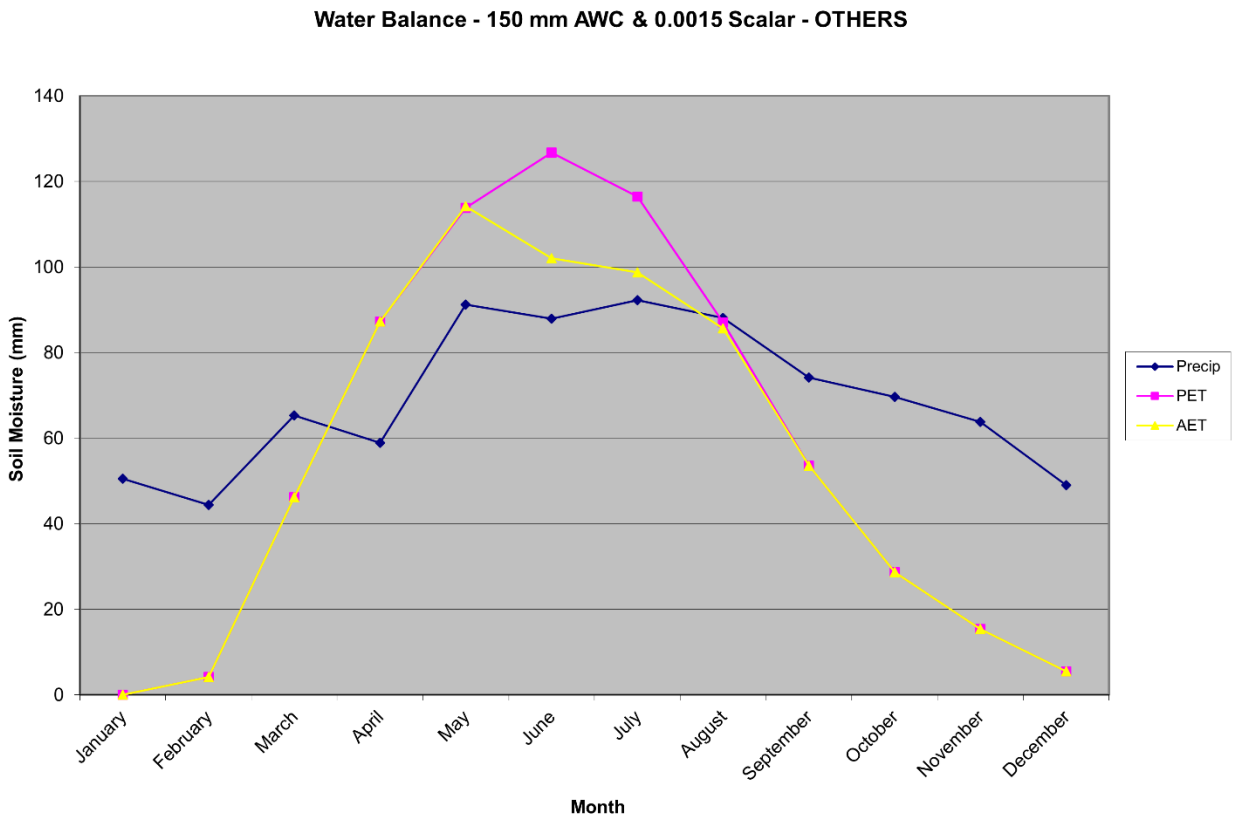

Figure A16. Graph showing monthly actual evapotranspiration (AET), potential evapotranspiration (PET), and precipitation (Precip) for OTHERS under the New multiplier scenario using an IMSI Scalar of 0.0015 in lieu of 0.0020 with the 150 mm soil available water capacity (AWC).